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// OSSP Skill 15 - Combined Redirection
// Demonstrates stdout + stderr merged into one file using dup2().

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main(void) {
    const char *file = "skill15_combined.txt";

    printf("=== Skill 15: Combined stdout + stderr Redirection ===\n");

    pid_t pid = fork();
    if (pid == -1) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        int fd = open(file, O_WRONLY | O_CREAT | O_TRUNC, 0644);
        if (fd == -1) {
            perror("open");
            _exit(1);
        }
    }
}
```

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sai_pavani@Sai-Pavani: ~/OS_ × + v
GNU nano 8.7.1 skill_15.c.c

    // Both streams point to the same open file description.
    dup2(fd, STDOUT_FILENO);
    dup2(fd, STDERR_FILENO);
    close(fd);

    printf("Normal output: stdout message.\n");
    fprintf(stderr, "Error output: stderr message.\n");

    fflush(stdout);
    fflush(stderr);
    _exit(0);
}

waitpid(pid, NULL, 0);

printf("Combined output written to %s\n", file);
printf("Reading the file:\n");

FILE *f = fopen(file, "r");
if (!f) {
    perror("fopen");
    return 1;
}

char line[256];
while (fgets(line, sizeof(line), f))
    printf("%s", line);
```

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    fclose(f);
    return 0;
}
```

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sai_pavani@Sai-Pavani:~/OS_SKILL$ nano skill_15.c.c
sai_pavani@Sai-Pavani:~/OS_SKILL$ gcc skill_15.c.c -o skill_15
sai_pavani@Sai-Pavani:~/OS_SKILL$ ./skill_15
=== Skill 15: Combined stdout + stderr Redirection ===
Combined output written to skill15_combined.txt
Reading the file:
Normal output: stdout message.
Error output: stderr message.
sai_pavani@Sai-Pavani:~/OS_SKILL$
```